BUU33803 Group Project, 2026-27

“Is the UK better or worse off after Brexit?”

Worth 50% of the module. Groups of four. Report, video presentation, live questions, and your Claude log.

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1. The question, and why you can’t answer it as written

“Better off” is not a number. Richer? Richer per person? Trading more? Investing more? More jobs, better-paid jobs, lower prices, lower emissions? Better off for whom: London, or Sunderland?

Your first job is to turn the question into one you can actually answer with data, and to say plainly what that choice leaves out. That is the same move as Session 1 (business question → data question), on a national scale.

Brexit brought a friend. The UK left the single market on 1 January 2021, in the middle of a pandemic that hit every country in the pack. Almost everything that changed in 2020–21 has two candidate causes. That is the newspaper-and-radio problem from Session 2, one scale up. A good report says what it can and cannot separate. It does not pretend.

> **An honest “the evidence points this way, this far, and no further” can earn a First. A confident verdict the analysis cannot support will not.**

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2. The dataset pack (everyone gets the same data)

Three files, all real, all public, frozen for the term. Download them from Blackboard.

File	What it is	Rows
`countries_quarterly_gdp.csv`	Real GDP (volume index) for the UK and 19 comparator economies, every quarter from 2010-Q1 to 2026-Q2. Rebased so **2016-Q2, the referendum quarter, = 100** for every country. Source: OECD Quarterly National Accounts.	1,320
`countries_annual.csv`	The same 20 countries, every year 2005–2024: GDP per head, GDP growth, trade, exports, investment, foreign direct investment, inflation, unemployment, CO2 per head. Source: World Bank, World Development Indicators.	400

File	What it is	Rows
`uk_areas.csv`	Every counting area in Great Britain in the 2016 referendum (380): the Leave vote, turnout, and what the area looked like before and after: employment rate (2015, 2019, 2024), share of adults with a degree-level qualification (2015), median full-time weekly pay (2015, 2024). Sources: Electoral Commission; ONS Annual Population Survey and ASHE via Nomis.	380

The 20 countries: United Kingdom; EU members France, Germany, Italy, Spain, Netherlands, Belgium, Austria, Denmark, Sweden, Finland, Ireland, Portugal; outside the EU United States, Canada, Japan, Australia, New Zealand, Norway, Switzerland.

Rules on data - Your core analysis must use the pack. That is what gets marked. - You may bring in extra data to support a point (trade with the EU specifically, for example). If you do: cite it, include the file with your submission, and say why the pack wasn't enough. - You may cite published estimates of the Brexit effect, and Claude knows several. But quoting someone else's number is not analysis. Marks come from what *you* found in *this* data, and from how you compare it with theirs.

Things in the pack that will catch you out if you don't look (we'll meet the ideas in class): - One of the comparators has GDP figures famously distorted by multinational accounting. Find it before it finds you. - The quarterly file is *total* GDP. Countries whose population grew fast look better on totals than per head. The annual file has GDP per head. Which one answers your question? - Pay in `uk_areas.csv` is in pounds of the day (nominal), not adjusted for inflation. - Some areas have blanks. Several English councils merged after 2019 and the 2024 pay series does not follow them. Decide what to do and say what you did.

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3. What your analysis must include

1. **A defined question.** One or two sentences: what "better off" means in your report, measured by which columns, over which years. Justify the choice, including what it misses.
2. **A comparison against the other countries, before and after.** Using regression (Sessions 2–3), compare the UK with a comparison group of your choosing, before and after the referendum and/or the end of the transition period. **You must state your comparison group, justify it, and show what happens to your answer if you choose a different one.** This is the heart of the project.
3. **Inside the UK, one more method from the course, on `uk_areas.csv`.** Choose one:
   - *Logistic regression* (Sessions 4–5): what did an area that voted Leave look like in 2015? How well can you predict it, and what does that tell you (and not tell you)?
   - *Clustering* (Sessions 6–7): group the areas, name the groups from what the data shows (not from assumptions about who lives there), and compare how each group fared after 2016.
4. **The confound, handled explicitly.** What did you do about Covid, and what can't you separate?
5. **Who and what.** At least one paragraph on who inside the UK is better or worse off (the social dimension) and one on the environmental dimension (CO_2 per head is in the pack). "Better off" need not mean "richer".

6. **A recommendation.** If a UK minister, an Irish exporter, or an investor asked “so what?”, what would you tell them, and how confident are you?

Python in Colab with Claude, as in class. Submit your notebook.

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4. Deliverables and deadlines

What	Detail	Due
Groups of four	Self-selected, via the Groups tab on Blackboard. Anyone not in a group by the deadline is placed in one.	Fri 2 Oct, 17:00
Question check-in (not graded)	One paragraph: your defined question, your comparison group, your inside-the-UK method. Feedback within a week.	Tue 13 Oct
Project clinic	Session 9: bring your draft analysis.	Tue 24 Nov
Video presentation	Max **8 minutes**. Every member appears and speaks. Findings, not a methods tour.	Mon 30 Nov, 17:00
Live questions	In Session 10, each group answers questions on its own work, **in person**, 3 minutes. Any member may be asked anything.	Tue 8 Dec
Report + notebook + Claude log	Report max **2,000 words** (tables, charts, appendix excluded). Group cover sheet with the AI statement.	Fri 11 Dec, 17:00

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5. How it is marked (50 marks)

Component	Marks	What earns the marks
Framing the question	7	A precise, measurable question; honest about what the choice leaves out.
Analysis	12	Sound use of regression and one other course method; comparison group justified and **tested** by changing it; the data traps noticed and handled.
Interpretation and the confound	10	Numbers read correctly; claims sized to the evidence; Covid dealt with explicitly; correlation not sold as cause.

Component	Marks	What earns the marks
Social, environmental and limitations	6	Who wins and loses inside the UK; the environmental angle; the limits of the data and the method.
Video presentation	6	Clear story, good charts, every member contributes, fits in 8 minutes.
Live questions	4	Every member can explain and defend the group's own choices.
Claude log and reflection	5	See section 6.

Every component is marked in four bands: **Fail / Pass / Merit / First**.

What separates a First from a Pass: a Pass report reads well and restates what is known about Brexit. A First report shows the reader something they could only learn from *this* analysis: how much the answer moves with the comparison group, where the Covid problem bites and where it doesn't, which areas the average hides. It is careful about what it cannot claim.

The live questions can move the report mark. If a group cannot explain its own analysis, the report mark is adjusted to match what the group can show it understands. This applies in both directions.

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6. Using Claude

Claude is required, as in class. It is also the easiest way to hand in a fluent, generic report that earns a Pass. Claude has read a great deal about Brexit, and if you ask it the question it will answer with other people's numbers. Use it to *do* your analysis, not to *be* your analysis.

Submit: - **The full Claude conversation log(s)** for the project (export, don't curate). - **A reflection, max 500 words**, with the group's name on it: one place Claude helped you think better, and **one place Claude was wrong, overconfident, or quietly steered you, and how you caught it.** A reflection with no example of catching Claude out cannot reach Merit.

The log is marked on the quality of your questions and how critically you treated the answers, the same standard as in class.

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7. Group work

- Groups of four. If a group falls to three, tell Eoin early. Scope stays the same.
- Include a short **contribution statement** in the appendix: who did what, signed by all four.
- If a group has a serious contribution problem, raise it before the deadline, not after.

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8. Keeping it civil

This is a politically charged question and people in the room will have strong views, some personal. The project asks *what the evidence supports*, not *whether Brexit was right*. You can reach any conclusion the analysis supports. You will lose marks for claims the analysis doesn't support, whichever side they favour.